

UNIVERSITY OF CÓRDOBA

MASTER'S FINAL PROJECT

Intelligent Analysis of the Application of Virtual Laboratories for Neurodevelopmental Disorders at the Early Childhood Intervention Center

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Abstract

The use of virtual laboratories has gained relevance as an innovative tool for intervention in neurodevelopmental disorders. This study focuses on the intelligent analysis of virtual laboratory applications at the University of Córdoba's Early Childhood Intervention Center, evaluating their effectiveness in therapy for children with neurodevelopmental disorders. By integrating artificial intelligence techniques, therapeutic session data is processed to identify patterns, measure progress, and optimize interventions.

The research uses a mixed-methods approach, combining qualitative observations with quantitative data analysis. Machine learning algorithms and statistical models are employed to assess the impact of virtual laboratories on cognitive, motor, and social development. Feedback from experts and families is also considered to provide a comprehensive perspective on the tool's effectiveness.

Preliminary findings suggest virtual laboratories can improve motivation, optimize learning outcomes, and personalize therapeutic approaches. However, key challenges remain, including accessibility, adaptation to individual needs, and the role of human interaction. This study contributes to the growing field of digital therapy, providing insights to enhance early interventions and guide future applications of virtual reality in neurodevelopmental therapy.

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1 Introduction

Neurodevelopmental disorders encompass a range of conditions that affect children's cognitive, behavioral, emotional, and social development. These disorders, with their varied manifestations, present significant challenges for the educational and therapeutic support of affected children.

Among these, Attention Deficit Hyperactivity Disorder (ADHD) is one of the most prevalent, with an estimated global prevalence of 8%, primarily affecting males. It is characterized by attention difficulties, impulsivity, and sometimes motor hyperactivity. The inattentive subtype is most common, and ADHD frequently co-occurs with other conditions such as Oppositional Defiant Disorder (ODD) or emotional disorders, complicating school and social interventions (Capobianco et al., 2025).

Specific Learning Disorders (SLD), including dyslexia, dyscalculia, and dysgraphia, lead to persistent difficulties in acquiring basic academic skills. These disorders are often linked to deficits in working memory or executive functions and frequently coexist with ADHD. Gender differences are observed: girls more commonly struggle with mathematics, while boys face challenges in spelling (Capobianco et al., 2025).

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition marked by difficulties in social interaction and communication, along with repetitive and stereotyped behaviors. While its exact causes remain unclear, research suggests genetic, environmental, and neurobiological factors. Diagnostic prevalence has risen sharply in recent decades, now affecting 1 in 77 children (Yang et al., 2025).

Other disorders, such as Selective Mutism (SM) or emotional regulation disorders, may also contribute to this clinical picture. Though sometimes secondary, these challenges impact children's overall development and interfere with learning.

Given the diversity and complexity of these disorders, immersive technologies like Virtual Reality (VR) are generating growing interest. Thanks to their interactive and immersive capabilities, VR offers controlled, customizable, and engaging environments for children. These features make it a promising therapeutic tool for targeting specific cognitive functions in a playful and motivating way.

Numerous studies now explore the use of VR-based games in treating neurodevelopmental disorders. Results often show short-term improvements, particularly in attention, memory, and emotional regulation. However, limitations persist, including a lack of longitudinal studies, cognitive fatigue-related side effects, and accessibility barriers for certain populations. Additionally, some disorder categories—particularly learning-specific ones—remain understudied in the scientific literature.

This state-of-the-art review therefore aims to examine existing research on VR games as tools for addressing cognitive functions in children with neurodevelopmental disorders. It identifies key findings, methodological limitations, and gaps in current literature. These insights will inform the development of a novel therapeutic VR game, whose efficacy will be evaluated through observed cognitive progress in children after use.

2 Neurodevelopmental Disorders (NDDs)

Neurodevelopmental disorders (NDDs) refer to a group of conditions that appear early in a child's life, usually before school entry. These disorders are caused by alterations in brain development, which can durably affect cognitive, emotional, behavioral, motor, and/or social functioning. They are defined by a deviation from or delay in typical development, with significant impacts on the individual's adaptive capacities (Morris-Rosendahl and Crocq, 2020).

NDDs often share common neurobiological foundations, which explains their high comorbidity (multiple disorders present in the same child) (Micheletti et al., 2025).

3 Classification of NDDs :

There are two main types of classification for mental disorders: on the one hand, the categorical approach, as exemplified by the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) and the International Classification of Diseases, 11th Revision (ICD-11), on the other hand, dimensional and biobehavioral approaches, such as the Hierarchical Taxonomy of Psychopathology (HiTOP) and the Research Domain Criteria (RDoC) (Tanaka, 2024).

3.1 Categorical Classification of NDDs

The Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) and the International Classification of Diseases, 11th Revision (ICD-11) are foundational tools in the field of mental health. These categorical classification systems provide clinicians with standardized criteria for diagnosing mental disorders. One of the main advantages of the categorical approach is that it facilitates clear and consistent communication among healthcare professionals. These manuals help ensure that practitioners across various settings and regions can accurately identify and treat mental health conditions.

However, as highlighted by (Tanaka, 2024), despite their widespread use, both the DSM-5 and ICD-11 have been subject to criticism for their limitations. The categorical model can sometimes oversimplify the complexity of mental health by forcing symptoms into rigid diagnostic categories. Issues such as comorbidity and heterogeneity within diagnostic groups underscore the need for a more dimensional understanding of mental disorders.

3.1.1 DSM-5 Classification

The DSM-5 (American Psychiatric Association, 2013) classifies NDDs into several specific categories based on standardized diagnostic criteria. It is predominantly used in clinical psychiatry and research, especially in North America.

The main DSM-5 categories include:

Intellectual Disability (ID)

Communication Disorders

Autism Spectrum Disorder (ASD)

Attention-Deficit/Hyperactivity Disorder (ADHD)

Specific Learning Disorder (SLD)

Motor Disorders (including Developmental Coordination Disorder and Tic Disorders)

Other Specified and Unspecified Neurodevelopmental Disorders

3.1.2 ICD-11 Classification

The World Health Organization's ICD-11 (2019) classifies NDDs within chapter 6A, aimed at global medical coding and epidemiological use. The corresponding ICD-11 codes align closely with DSM-5 categories, facilitating international harmonization.

ICD-11 NDD categories include:

6A00 – Intellectual Developmental Disorders

6A01 – Developmental Speech or Language Disorders

6A02 – Autism Spectrum Disorder

6A03 – Attention Deficit Hyperactivity Disorder

6A04 – Developmental Learning Disorder

6A05 – Developmental Motor Coordination Disorder

6A0Y/6A0Z – Other or Unspecified Neurodevelopmental Disorders

Within the framework of categorical classification, one of the primary challenges in diagnosing neurodevelopmental disorders (NDDs) lies in the high prevalence of comorbidity. As reported by (Francés et al., 2023), a substantial proportion of six-year-old children diagnosed with an NDD were found to have multiple concurrent diagnoses. Specifically, 76.5% of the assessed children presented with between two and six co-occurring NDDs, calling into question the validity and utility of strictly delineated diagnostic categories.

The most frequently observed comorbid patterns included attention-deficit/hyperactivity disorder (ADHD) alongside specific learning disorders (in reading and/or writing), the co-occurrence of communication and learning disorders, and complex clinical profiles involving ADHD, language impairments, and cognitive difficulties.

Moreover, nearly 89% of the diagnosed children had not previously been identified or referred for any neurodevelopmental concern, underscoring both the under-detection of NDDs and the limitations of relying solely on categorical criteria for early identification.

These findings suggest that comorbidity is not an exception but a structural characteristic of NDDs. They highlight the constraints of categorical classification systems, which may oversimplify or obscure the complexity of neurodevelopmental profiles. Consequently, there is a critical need for integrative and multidimensional assessment approaches that can account for overlapping symptoms and provide a more accurate foundation for diagnosis and intervention.

3.2 Dimensional Classification of NDDs

Dimensional classification is an approach that considers the symptoms of mental and neurodevelopmental disorders as traits that exist on a continuum, rather than as distinct categories. Instead of saying that a disorder is either “present” or “absent,” this approach allows us to measure the intensity of difficulties and monitor how they change over time (Ogundele and Morton, 2022; Frazier et al., 2023). It highlights the diversity of individual profiles and helps tailor care to each child’s specific needs.

Until now, most diagnostic systems, such as the *DSM-5* or the *ICD-11*, have used a categorical approach. This means a diagnosis is made only if a child shows a certain number of well-defined symptoms. It’s a binary model: the child either has the disorder or does not.

However, this method has several limitations. In real life, many children experience significant difficulties without meeting all the official criteria for a diagnosis. These so-called “sub-threshold” cases can still have a major impact on their daily lives (Ogundele and Morton, 2022). Moreover, symptoms from different disorders often overlap—for example, a child may show signs of autism, anxiety, or language difficulties all at once—making classification more difficult.

Dimensional approaches aim to overcome these limitations by offering more flexible and precise assessments. They allow clinicians to measure symptoms on a scale, helping

to understand how severe the difficulties are and how they evolve. This supports more personalized care, even for children who do not meet full diagnostic criteria but still need support.

This reflection on the limitations of categorical approaches and the contributions of dimensional approaches resonates particularly with two major contemporary models: HiTOP (Hierarchical Taxonomy of Psychopathology) and RDoC (Research Domain Criteria).

3.2.1 Hierarchical Taxonomy of Psychopathology (HiTOP)

The Hierarchical Taxonomy of Psychopathology (HiTOP) represents a major advancement compared to traditional categorical models of mental disorder classification. Rather than viewing mental disorders as distinct entities, HiTOP organizes psychopathology along a dimensional spectrum and within a hierarchical structure. This approach accounts for the overlap of symptoms as well as the comorbidity frequently observed in mental health diagnoses. By enabling more personalized assessments and interventions, HiTOP aims to improve diagnostic accuracy and, potentially, the effectiveness of treatments.

The HiTOP model (Hierarchical Taxonomy of Psychopathology) is built on a hierarchical structure that organizes psychopathological manifestations across multiple levels of abstraction, ranging from specific symptoms to a general factor of psychopathology.

Level 1: Symptoms and Traits At the base of the hierarchy are the elementary components of psychopathology, including both clinical symptoms (e.g., insomnia, rumination, avoidance behavior) and maladaptive personality traits (e.g., impulsivity, hostility). These components may be transient (symptoms) or stable (traits), and represent the foundational units of assessment.

Level 2: Syndromes Symptoms and traits are grouped into dimensional syndromes, defined as coherent clusters of clinically interrelated manifestations, empirically identified through covariation. Unlike traditional diagnoses, these syndromes are not based on arbitrary thresholds but on continuous latent structures (e.g., social anxiety, depression, alcohol use disorder).

Level 3: Subfactors Syndromes are further aggregated into subfactors, representing intermediate clinical groupings with shared features. Examples include:

- Fear (e.g., phobias, panic disorder),
- Distress (e.g., major depression, generalized anxiety),
- Antisocial behavior (e.g., conduct disorder, oppositional defiant disorder).

Level 4: Spectra Subfactors are integrated into broad psychopathological spectra,

which capture transdiagnostic dimensions of mental illness. Six major spectra have been validated so far:

- Internalizing (e.g., mood and anxiety disorders),
- Externalizing – disinhibited (e.g., impulsivity, substance use),
- Externalizing – antagonistic (e.g., hostility, manipulation),
- Thought disorder (e.g., psychotic symptoms),
- Somatoform (e.g., medically unexplained physical symptoms),
- Eating and sexual problems (currently partially integrated).

Level 5: Superspectra and General Factor Spectra may themselves be grouped into superspectra, such as emotional dysfunction, broad externalizing, or psychosis (some of which remain provisional). At the top of the hierarchy lies the general factor of psychopathology, known as the p factor, which reflects a shared vulnerability across all forms of mental illness. This construct is conceptually analogous to the general intelligence factor (g factor) in cognitive psychology and helps account for pervasive comorbidity and overall psychological dysfunction.

Clinical and Scientific Relevance This hierarchical structure enables a more nuanced and integrative understanding of psychopathology. It facilitates the identification of transdiagnostic patterns, improves diagnostic reliability, and enhances the external validity of clinical constructs. From a clinical perspective, HiTOP supports a more individualized assessment and treatment approach by targeting broad and specific dimensions rather than rigid diagnostic categories. (Kotov et al., 2021)

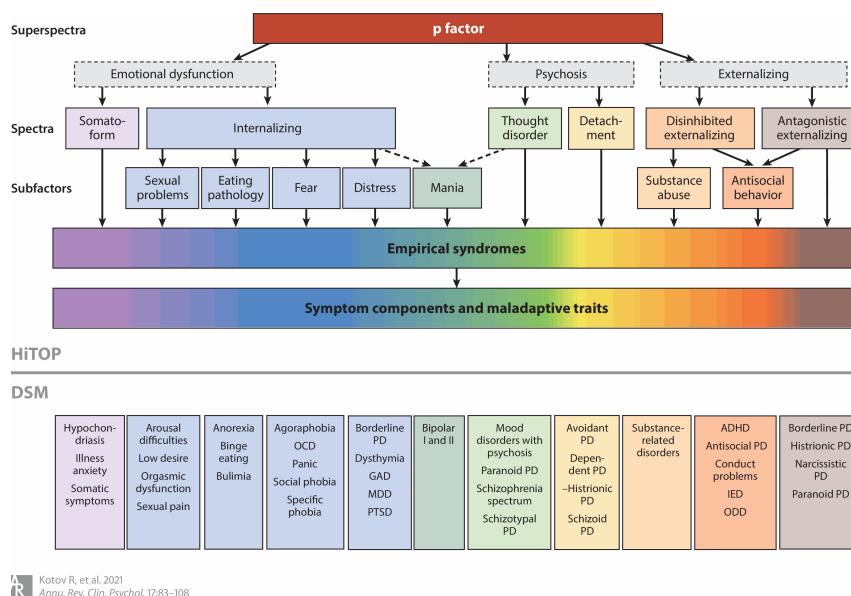


Figure 1: HiTOP model.

3.2.2 Research Domain Criteria (RDoC)

The Research Domain Criteria (RDoC) model, introduced by the National Institute of Mental Health (NIMH) in 2009, offers an alternative conceptual framework for research in psychopathology. In contrast to traditional approaches based on categorical diagnostic systems such as the DSM, the RDoC model adopts a dimensional and transdiagnostic perspective centered on the core mechanisms underlying mental functioning. Its primary objective is to enhance the understanding of the biological, cognitive, and behavioral bases of mental disorders, across the full spectrum from typical to atypical functioning.

The RDoC framework is based on the premise that psychiatric disorders result from dysfunctions in fundamental psychological and neurobiological systems. These systems are conceptualized as functional domains, each reflecting a core aspect of human behavior and its biological underpinnings. For instance, domains include negative valence systems (associated with threat or loss), positive valence systems (involving reward and motivation), cognitive systems (related to executive functions, attention, and memory), social processes (such as the perception and understanding of others), arousal and regulatory systems, and sensorimotor systems.

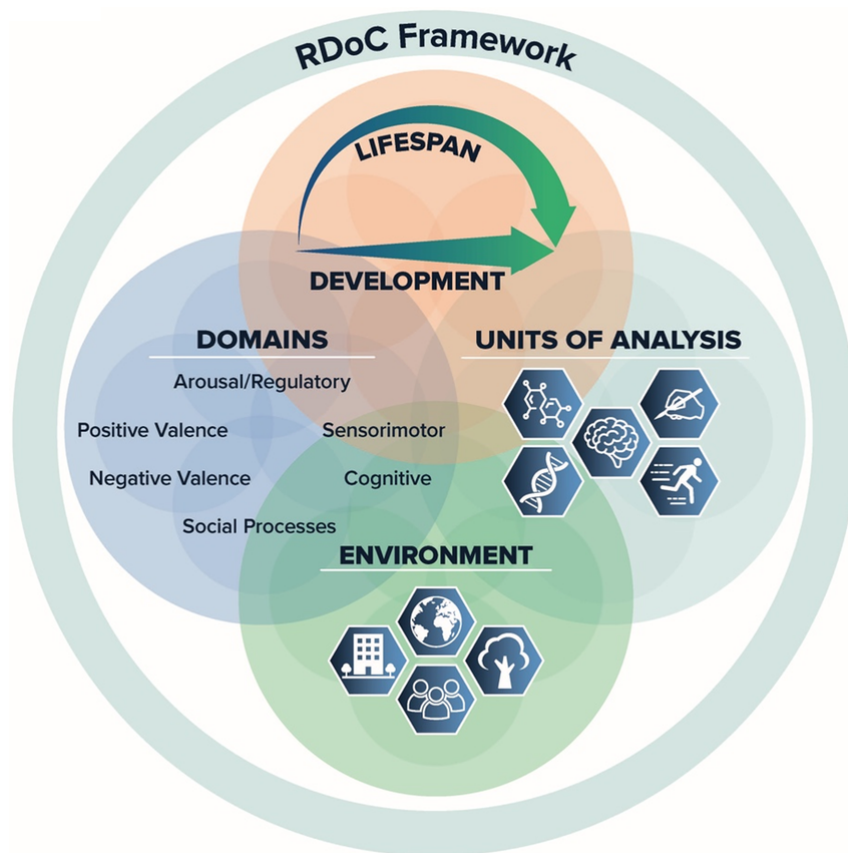


Figure 2: RDoC model.

Each domain is examined through multiple levels of analysis, encompassing genetics, molecular and cellular biology, neural circuits, physiological measures, observable behavior, and self-reported experiences. The intersection of these domains and units of analysis forms the RDoC matrix, which serves as a flexible tool for hypothesis generation and the organization of empirical data.

The RDoC matrix is designed to be an evolving, non-prescriptive resource. It is not intended to constrain research within a fixed set of categories, but rather to provide a dynamic structure capable of integrating new constructs or levels of analysis as scientific knowledge advances. Although developmental processes are not represented as a standalone domain in the matrix, the RDoC framework explicitly acknowledges their cross-cutting importance. Developmental trajectories and interactions with environmental factors are regarded as essential dimensions in the study of variations in mental functioning across the lifespan.

The RDoC approach addresses several limitations inherent to traditional diagnostic frameworks, such as intra-diagnostic heterogeneity and symptom overlap across disorders.

It provides a relevant structure for investigating mechanisms that cut across diagnostic categories, clarifying within-disorder variability, and identifying early and personalized targets for intervention. This model has proven particularly valuable in research on neurodevelopmental disorders in children and adolescents, facilitating the investigation of shared underlying mechanisms across seemingly distinct clinical presentations (Pacheco et al., 2022).

System	Descriptions	Strengths	Weaknesses
DSM-5	Primarily used in the USA for clinical diagnosis and research	Provides detailed criteria for diagnosis, widely used in research settings, and has a long history of use	Criticized for lack of validity, influenced by commercial factors, and slow to incorporate new findings
ICD-11	Used globally for clinical diagnosis and public health purposes	Harmonized with DSM-5 to some extent, focuses on clinical utility, and is widely accepted internationally	Contains some disorder categories not present in DSM-5, and differences in priorities and uses can lead to inconsistencies
HiTOP	Classify mental disorders based on empirical data and dimensional traits	Integrates maladaptive personality traits into a single system, offers a dimensional approach that may better capture the complexity of mental disorders	Still under development and less widely adopted compared to DSM-5 and ICD-11
RDoC	Integrate basic behavioral and neuroscience research to understand mental disorders	Focuses on understanding the biological bases of mental disorders, offers a dimensional approach that can enhance research precision	Lacks extensive validation, and its practical application in clinical settings is still limited

Table 1: Summary table of mental disorder classification systems

4 Virtual Reality as an Intervention Tool

Virtual reality (VR) constitutes an immersive technology capable of simulating interactive 3D environments with real-time user navigation and interaction. The system employs VR headsets, motion controllers, and body tracking platforms to generate comprehensive sensory immersion.

In the field of health and education, VR is increasingly used as an intervention tool, especially with children who have neurodevelopmental disorders. It provides a controlled, repetitive and adaptable environment that facilitates learning of cognitive, social or behavioral skills. Thanks to the immersion it provides, VR can capture the child's attention, reduce external distractions and promote engagement in specific tasks.

VR can be used in non-immersive way (on conventional screens), semi-immersive (3D screens or projectors), or fully immersive (VR headset). These levels of immersion influence the quality of the experience and the potential effects on learning and behavior.

Numerous previous studies have focused on the relevance and effectiveness of virtual reality in the context of diagnosis and treatment of children {Capobianco2025. Some research has also explored the ability of virtual reality interventions to alleviate symptoms such as aggressiveness, anger and impulsivity, while questioning the potential of immersion as an alternative or complement to conventional therapeutic approaches (Roncero et al., 2025). On the other hand, other works have been interested in the possible side effects associated with the use of virtual reality (Yang et al., 2025).

Furthermore, several investigations have focused on the set of serious games, whether immersive or non-immersive, in order to evaluate their effectiveness in improving executive functions, as well as metacognitive and emotional competencies in children with attention deficit disorder with or without hyperactivity (ADHD) (Doulou et al., 2025).

It is important to note that children suffering from ADHD, developmental coordination disorder (DCD), or presenting comorbidity between both disorders, often show significant motor deficits. However, current interventions tend to focus on final learning outcomes, neglecting detailed analysis of the learning process itself (Cai et al., 2025).

In this research, we will specifically focus on the use of immersive virtual reality as an intervention tool with children who have neurodevelopmental disorders.

4.1 Immersive VR Applications in School Environments

Studies using immersive virtual reality show significant improvements in children with ADHD, particularly in behavioral flexibility, attention capacity, motor self-regulation, and

planning.

In the school context, a scientific review analyzed five investigations that directly compared a virtual classroom with a real classroom. The virtual environment faithfully reproduced a typical classroom, with rows of desks, a blackboard, doors, a window with a view of the playground, as well as various sources of visual distraction (people entering and leaving, paper airplanes) and auditory distraction (chair noises, whispers, objects falling to the ground).

The use of virtual reality headsets allows minimizing external sensory stimuli, thus promoting total immersion in the simulated environment. This immersion, combined with tasks focused on attention and inhibition of impulsive responses, allowed identifying different cognitive profiles in children with ADHD. Additionally, the presence of an avatar, such as a virtual teacher, appears to facilitate the use of immersive VR and improve academic performance in this context (Capobianco et al., 2025).

Among the studies included in the review, only those published between 2020 and 2025 were selected (Table 1)

Disease / Age Range and N	Geographic Origin	Outcomes	Limitations
ADHD / 7-18y (N=164)	China, USA, South Korea, Iran	VR for global cognitive functioning, attention, and memory	Poor quality of the studies, which could affect validity
ADHD / 6-12y (N=90)	China	VR in ADHD children who struggle with real-world social settings	Results from Hong Kong may not be generalizable to other contexts or populations
ADHD / 6-12y (N=45)	USA	VR and decision-making in individuals with ADHD	Small sample size may limit the generalizability of results
ADHD / 9-12y (N=38)	Finland	VR as an alternative methodology	Sample size restricts analytical techniques
ADHD / 6-16y (N=37)	Taiwan	VR as a free alternative to traditional techniques	Relatively small sample size
ADHD / 9-13y (N=37)	Finland	VR seen as non-invasive for ADHD participants	Generalizability of results is a major concern

Disease / Age Range and N	Geographic Origin	Outcomes	Limitations
ADHD / 12-22y (N=37)	Spain	VR use increased engagement and motivation	Limited data on ADHD; findings mostly from other mental health contexts
ADHD / 6-12y (N=25)	China	VR enables more effective therapeutic experiences than traditional methods	High cost and limited availability of VR systems in some settings
ADHD / 9-17y (N=21)	South Korea	VR as a tool to study the impact of distractors on ADHD behavior	Small sample size limits generalizability
ADHD / 8-12y (N=20)	USA	VR precision could enhance understanding of ADHD mechanisms	VR systems may be expensive and not widely available
ADHD / Not specified (N=?)	Indonesia	VR offers a unique and stimulating context for behavioral assessment	Advanced VR systems require significant investment

Table 2: Summary of studies using virtual reality in children and adolescents with ADHD

The table (Table 1) summarizes several studies using virtual reality with children and adolescents with ADHD. These studies, conducted in various countries, show positive effects on attention, memory, motivation, and decision-making. However, they present recurring limitations, such as small sample sizes, limited generalizability of results, and constraints in accessing the technology. Despite this, VR appears as a promising tool for assessment and intervention in the context of ADHD.

The review (Capobianco et al., 2025), based on an analysis of several recent studies, highlights the potential of virtual reality (VR) as an assessment and intervention tool for children with ADHD. It shows that virtual environments, particularly school classroom simulations, allow for more precise identification of typical manifestations of the disorder - such as inattention, impulsivity, and hyperactivity - while providing an ecologically valid framework. Moreover, VR would promote emotional engagement, motivation, and reduced anxiety, thereby improving children's performance during tasks. The review also emphasizes the interest of complementary approaches, such as the use of immersive telemedicine or eye-tracking technologies, in a screening and therapeutic support approach. These re-

sults suggest that an integrated treatment, combining virtual immersion with cognitive or behavioral objectives, could enhance intervention effectiveness for this population.

Research indicates that children with attention deficit hyperactivity disorder (ADHD) commonly experience comorbid specific learning disorders (SLDs), particularly in academic contexts where attention deficits significantly impact learning outcomes (Capobianco et al., 2025). However, as identified in the current literature review, there remains a notable gap in studies examining virtual reality (VR)-based interventions, whether immersive or non-immersive, for this comorbid population (see Table 2).

Disease / Age Range and N	Geographic Origin	Outcomes	Limitations
ADHD & ASD / 8-16y N=76	USA	The VR paradigm could be a innovation over traditional method	It is impossible to determine if the VR job triggers, cognitive, social, or a combination of demands

Table 3: Summary of studies using virtual reality in children and adolescents with ADHD & ASD

4.2 Immersive Serious Games Designed for Neurodevelopmental Disorders

Serious games based on virtual reality (VR) technologies represent innovative therapeutic tools designed to support children with neurodevelopmental disorders (NDD), including attention-deficit/hyperactivity disorder (ADHD), autism spectrum disorders (ASD), and specific learning disorders (SLD). These immersive and engaging environments are developed to target specific cognitive functions, with the aim of fostering their development or rehabilitation.

The design of such serious games follows a structured methodological process comprising three core stages: the precise definition of therapeutic and educational objectives, the development of functional prototypes, and the rigorous evaluation of both usability and clinical effectiveness.

This evaluation phase encompasses several key components: the systematic collection of user feedback to refine the interactive experience; the identification and resolution of technical issues, accompanied by the optimization of game mechanics; and the validation of the system’s acceptability, ensuring that it is engaging, intuitive, and free of significant adverse effects (Rodrigo-Yanguas et al., 2021).

4.2.1 The Secret Trail of Moon (TSTM)

TSTM is a virtual reality-based game developed for children diagnosed with ADHD. The intervention comprises five distinct mini-games, each designed to train specific executive functions including sustained attention, working memory, cognitive flexibility, planning, and reasoning. Grounded in established cognitive models (Brown, Baddeley), these therapeutic activities aim to enhance impaired cognitive domains while maintaining high engagement through immersive gameplay (Rodrigo-Yanguas et al., 2021).

Smasher: In this minigame, participants must break a rock blocking their path by correctly following a sequence of chess pieces. The task is designed to assess two key cognitive skills:

Sustained attention (based on Brown's ADHD model), required to maintain focus on the sequence over time.

Inhibitory control, needed to resist impulsive moves and follow the rules accurately.



Figure 3: screenshot of the mini game Smasher.

Enigma: Participants must memorize the associations among different elements. Afterward, they must match the association as quickly as possible. The task is designed to assess two key cognitive skills:

- Working memory
- Cognitive flexibility

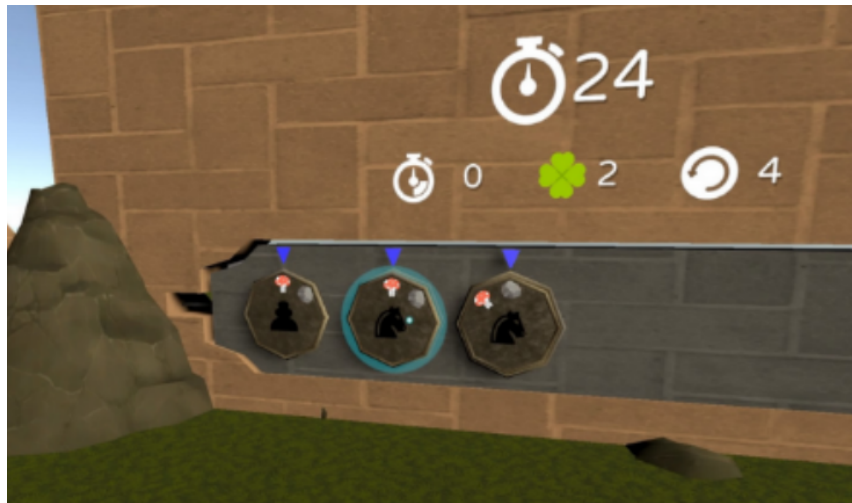


Figure 4: screenshot of the mini game Enigma.

Kuburi: Participants must create a drawing by using the face and orienting some cubes. The task is designed to assess two key cognitive skills:

Visuospatial ability According to the classic Baddeley model (1992), the visuospatial agenda is part of working memory.

Cognitive flexibility

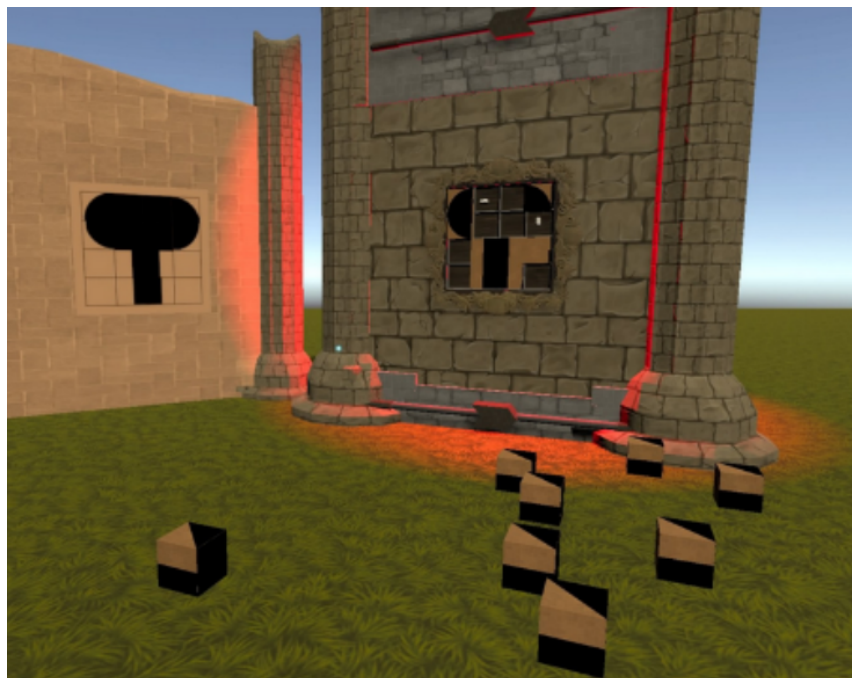


Figure 5: screenshot of the mini game Kuburi.

Teka Teki: Participants can obtain a key if they can help the fox follow her path to the lock. However, the path is obstructed by different blocks. The number of possible movements will decrease, thus increasing the difficulty. The task is designed to assess two key cognitive skills: Planning game

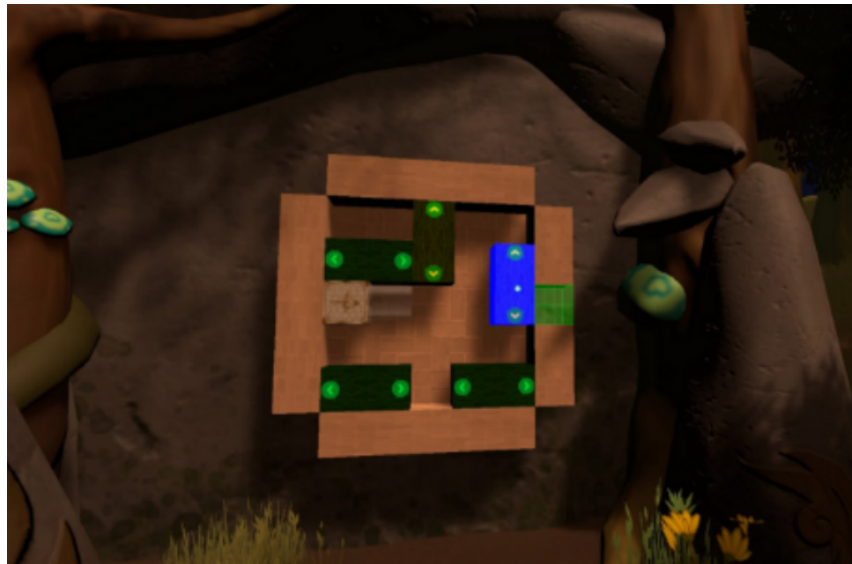


Figure 6: screenshot of the mini game Teka Teki.

Chessi: Participants must learn the basic rules of chess (ie, the movement of pieces, value of pieces, most relevant moves, etc). The level of difficulty progressively increases. The task is designed to assess two key cognitive skills: Reasoning

Planning

Math calculations



Figure 7: screenshot of the mini game Chess.

Effectiveness of the game: A randomized clinical trial, conducted between May and December 2023 in Spain, evaluated the effectiveness of a serious virtual reality video game called The Secret Trail of Moon (MOON) in improving emotional regulation in 76 children and adolescents with ADHD (attention deficit hyperactivity disorder). The participants, aged 7 to 18 and on stable medication, were divided into two groups: one received the usual treatment supplemented with 20 MOON gaming sessions, while the other received only standard treatment and psychoeducational support. The single-center, non-blinded study used validated questionnaires (such as the SDQ) to assess effects before, during, and after the intervention. Although the primary hypothesis of a significant improvement in emotional regulation was not confirmed, notable benefits were observed in working memory, material organization, and inhibition among the most engaged participants with the game (Martin-Moratinos et al., 2025).

4.2.2 Fruit Pioneer

Fruit Pioneer is an immersive serious game developed in virtual reality (VR) with the purpose of enhancing specific cognitive functions through an engaging and interactive task. Designed following principles from cognitive science and neuroplasticity, the game offers a structured cognitive training environment that maintains user motivation while targeting working memory, executive function, attention, and visuo-motor coordination.



Figure 8: Player's first-person view in the VR environment of. *Fruit Pioneer*.

The player, equipped with a VR headset (HTC Vive) and two handheld controllers, enters a virtual environment where the main task is to slice fruits that are launched from multiple directions. The action is performed using virtual blades that respond to the motion of the controllers. Simultaneously, the player must avoid iron balls that act as distractors and cognitive interference.



Figure 9: Virtual knives controlled by the user through hand-held devices

Each game session lasts two minutes, during which fruits appear with increasing frequency, speed, and in more dispersed trajectories. The player receives real-time feedback via a score counter and leaderboard, which enhances engagement and encourages performance improvement over multiple sessions.

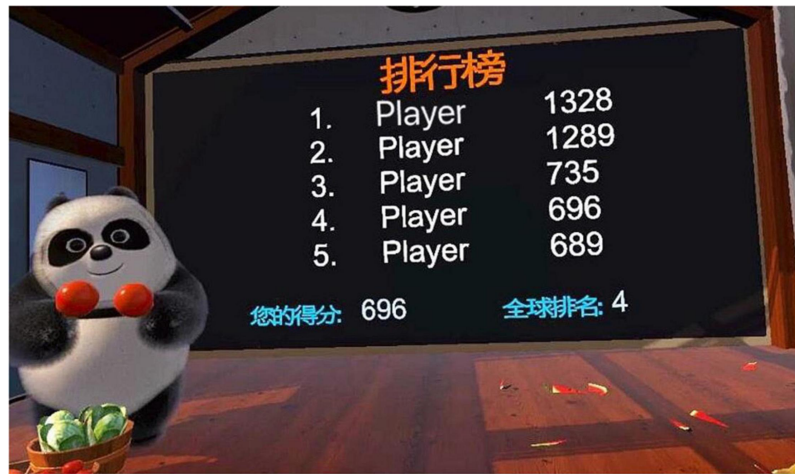


Figure 10: Leaderboard showing the player's score and ranking.

Fruit Pioneer specifically targets several cognitive domains. Working memory is trained through the need to remember which fruits yield the highest score and adapt one's strategy accordingly. Executive functions are engaged via processes of planning, inhibition (avoiding distractors), cognitive flexibility, and rapid decision-making. The immersive setting enhances sustained attention and reduces external distractions.

The game's design incorporates progressive difficulty and real-time interaction, which are known to support cognitive engagement. Immediate feedback and game-based motivation promote repeated training, a key factor in activating neuroplastic mechanisms.

By combining immersive interaction with specific cognitive challenges, *Fruit Pioneer* represents a promising tool for cognitive stimulation or rehabilitation. Its adaptability and standardization make it suitable for various clinical and research contexts, especially for populations with cognitive impairments due to psychiatric or neurological conditions. Although not tailored to a specific disorder, the game is particularly effective in addressing transdiagnostic cognitive symptoms such as reduced attention, poor inhibitory control, and deficits in working memory (Wang et al., 2022).

5 Methodology

5.1 Bibliographic Search Methodology

The first step consisted in clearly defining the scope of the research by formulating the central question: "How do existing interventions address the complexity of comorbid clinical profiles, and what are their limitations, particularly in terms of ecological validity and skill generalization?" This question guided the identification of keywords and the criteria for source selection. The objective was to identify innovative approaches, such as transdiag-

nostic and ecological interventions, that could inspire the development of our own solution.

In this context, the literature review was conducted systematically using keywords directly related to the project, such as “neurodevelopmental disorders virtual reality,” “autism virtual reality,” or “VR ADHD therapy.” The results were filtered based on titles, abstracts, and keywords, and the selected articles were then downloaded and organized in Mendeley, which enabled their classification by theme and year of publication.

To ensure the reliability of the sources, the search was carried out in well-established academic databases (Scopus, PubMed, ScienceDirect). Particular attention was given to journal quality (impact factor, peer review), author reputation, and citation count, with a preference for articles that had already gained significant recognition within the scientific community. At this stage of the analysis, 36 articles published between 2020 and 2025 have already been selected.

5.2 Methodological Approach and Objectives

The project is part of an interdisciplinary approach combining clinical psychology, cognitive science, and immersive technologies. The adopted methodology is based on a trans-diagnostic approach, aiming to treat symptoms directly rather than being limited to a categorical classification, and on an ecological grounding to ensure the transferability of acquired skills to children's daily lives.

This approach was guided by dimensional classifications (HiTOP and RDoC models), making it possible to account for the diversity of clinical profiles and the potential presence of comorbidities. The methodology also includes co-design with users and experts to maximize the intervention's suitability and effectiveness.

Primary objective: To design and develop a flexible and customizable intervention protocol capable of addressing a variety of co-occurring difficulties and promoting an optimal transfer of learning to daily life.

Secondary objectives:

To integrate techniques from different therapeutic frameworks (CBT, ACT, schema therapy).

To actively involve the child, and where relevant their close circle, in the co-construction of goals.

To use situations and tools that are close to the participants' daily reality to enhance ecological validity.

This methodological structure served as the foundation for the design and development of the “My Morning Adventure” prototype, ensuring consistency between the theory, pedagogical objectives, and practical elements of the intervention.

5.3 Prototype / Intervention Development

5.3.1 Tools and Resources Used

The prototype was developed using the Spatial.io and Sketchfab platforms on a Windows 10 laptop. For the testing phases, the developer used a VR headset and controllers. The children played on their mobile phones by activating the VR option, which allowed them to move the camera through motion. Pictograms from ARASAAC were integrated to guide the game's activities.

5.3.2 Implementation Steps

The prototype's design was based on dimensional classifications (HiTOP and RDoC), aiming to address symptoms directly and reduce problems related to comorbidity.

The design phase included the creation of use case diagrams and activity diagrams, followed by the development of the "My Morning Adventure" game, which replicates children's daily activities: packing the backpack, choosing food for breakfast, leaving the house, crossing the road, taking the bus to school, arriving in the classroom, and choosing a seat.

This phase lasted approximately two months, translating the theoretical objectives (transdiagnostic approach and ecological validity) into concrete, child-friendly interactive activities.

5.3.3 Testing and Adjustments

The evaluation of the prototype was conducted with a sample of 20 child-parent dyads. Participants' questionnaire responses were first organized in Google Sheets and then exported as CSV files, which facilitated statistical analysis using Python within the Google Colab environment. This workflow ensured both traceability and reproducibility of the analyses.

The testing sessions were carried out under the supervision of a clinical psychologist specialized in supporting children with neurodevelopmental disorders (NDDs). This professional oversight provided a safe and adapted framework, ensured qualitative observation of the children's interactions, and guaranteed that the evaluation respected the specific needs of the participants.

Analytical techniques applied

Descriptive statistics: means and standard deviations were calculated to capture central tendencies of responses.

Internal consistency analysis: Cronbach's alpha was computed to assess the reliability of the questionnaire scales.

Graphical visualization: histograms and charts were used to illustrate the distribution of responses from both children and parents.

Child–parent correlations: Pearson's correlation test was applied to evaluate the degree of association between child and parent responses.

Comparative analyses: mean scores were compared across disorder groups to explore potential differences between subpopulations.

Methodological adjustments included excluding Question 1 (Q1) from children's indices, since none of the children had used a VR headset. This decision prevented biased results and preserved the ecological validity of the prototype.

6 Game Scenario and Detailed Design

The "My Morning Adventure" prototype was designed to replicate a child's daily activities and incorporate targeted therapeutic goals. Each scenario concretely translates the project's theoretical objectives, using a transdiagnostic approach and ecological grounding.

6.1 Activity Overview

Activity 1: Packing the Backpack The player must follow the instructions provided by the pictograms to pack their backpack: take the book, the notebook, and choose their pencil case. This activity aims to strengthen organization, planning, and sequential memory skills.

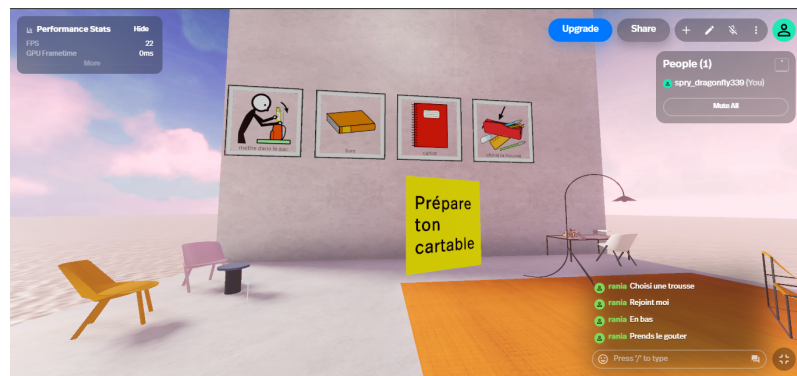


Figure 11: Packing the Backpack.



Figure 12: Packing the Backpack.

Activity 2: Having Breakfast The player is presented with a choice between an apple (a healthy food) and a cookie (a temptation to avoid). Each choice carries consequences, allowing the activity to focus on decision-making, inhibition, and understanding consequences.



Figure 13: Having Breakfast.

Activity 3: Leaving the House and Crossing the Road The player must find the exit door and follow the pictograms to cross the road safely: green traffic light, character looking left and right, then character crossing. This activity targets attention, motor planning, and road safety awareness.



Figure 14: Leaving the House.



Figure 15: Crossing the Road.

Activity 4: Taking the Bus and Arriving at School The player goes to the bus stop, boards the bus, and arrives near the school. Pictograms guide the player to the school (a character looking right and a school pictogram). This activity reinforces spatial orientation and understanding event sequences in daily life.



Figure 16: Taking the Bus.

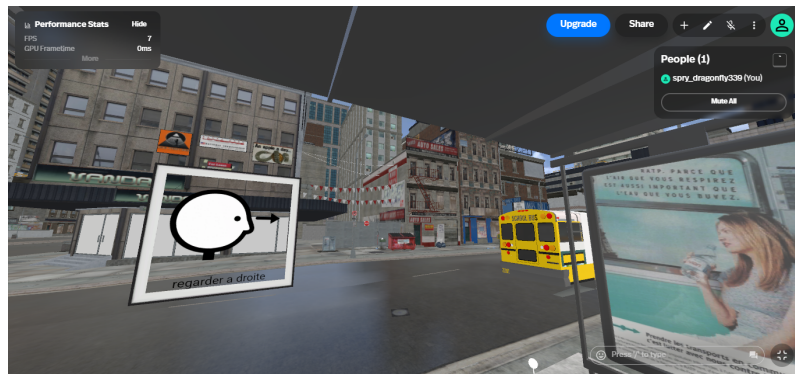


Figure 17: Arriving at School.



Figure 18: Arriving at School.

Activity 5: Choosing a Seat in the Classroom Upon arrival, the child must choose a seat in the classroom, which helps develop autonomy and decision-making skills in a social context.



Figure 19: Classroom.

6.2 Therapeutic Goals and Targeted Symptoms

Organization and Planning: Activities 1 and 3, through following pictograms and completing action sequences.

Attention and Inhibition: Activities 2 and 3, by presenting the player with choices and situations requiring impulse control.

Spatial Orientation and Navigation: Activities 3 and 4, to reinforce the ability to navigate everyday environments.

Autonomy and Decision-Making: Activities 2 and 5, by providing concrete choices and illustrating the consequences of each decision.

7 Discussion and evaluation

7.1 The clinical evaluation

of the prototype conducted by a clinical psychologist specialized in supporting children with neurodevelopmental disorders (NDDs), highlighted several strengths as well as important limitations to consider.

From a clinical perspective, the application represents a relevant external support tool to help structure daily routines. By adopting a dimensional approach, Mrs. Djouamaa identified three main areas of difficulty that the game specifically addresses:

Organization and planning: through the structuring of missions and transitional spaces, the child learns to sequence and organize actions in a logical order.

Impulsivity and executive functions: the use of explicit visual instructions encourages reflection before action, supporting the inhibition of impulsive behaviors.

Attention and concentration: clear pictograms and defined objectives, combined with tasks broken into small steps, help channel attention, maintain cognitive effort, and make the experience more rewarding.

She also emphasized the value of flexibility and choice offered to the player (e.g., deciding the order of missions, choosing between a healthy or less healthy snack, selecting a seat in class). These structured choices promote autonomy and decision-making within a safe environment.

However, several limitations were identified:

The limited number of tasks, which does not cover the full range of daily routines relevant to the child's age and needs.

The absence of explicit validation at the end of missions, reducing motivation and the reinforcement effect.

The lack of global structuring, as there is no initial checklist or clear transition to the next step.

Partial coherence of tasks, such as the schoolbag mission, which does not extend to carrying it to school. This reduces real-life transfer and educational impact.

7.2 Technical Justification

From a technical standpoint, these limitations are partly explained by the initial design choices, which focused on developing a functional prototype without full interactivity. The primary goal was to demonstrate feasibility and to translate theoretical objectives (transdiagnostic approach, ecological validity) into concrete elements. This explains the absence of automated feedback and the limited capacity to integrate complex actions such as carrying the schoolbag or validating the completion of missions.

Nevertheless, from the design phase onward, particular attention was given to the involvement of psychologists and parents. Their role was to accompany the child, provide explicit feedback, and reinforce the experience through human mediation. This choice reflects both the flexibility of the prototype (as a tool to be used with adult guidance) and its current limitations (which require more advanced automation in future iterations).

7.3 Analysis of Results

In order to better understand the perceptions and experiences of children and their parents, two separate questionnaires were administered. Each questionnaire included ten items (Q1–Q10), covering different dimensions such as usability, learning, engagement, and autonomy. Based on these responses, indices were constructed to synthesize the results.

Domain evaluated	Children (Questions)	Parents (Questions)
Usability	Q2: I felt comfortable while using the app. Q4: It was easy to follow the steps in the app. Q5: I could use the app without an adult's help.	Q2: Using the app was easy for my child/student. Q4: The pictograms and steps were easy for them to understand.
Learning	Q3: I understood the pictures (pictograms) well. Q6: Now I know better how to do the task the app taught me. Q7: I can remember the steps without looking at the app. Q8: The app helps me do the task at home or at school.	Q4: The pictograms and steps were easy for them to understand. Q5: After using the app, I noticed they understood the daily task better. Q6: I noticed they remembered the steps better after using the app. Q8: The app is useful for reinforcing learning of daily routines.
Engagement	Q1: I liked using the virtual reality headset. (<i>Note: no VR headset was used</i>) Q9: I would like to use this app again another day. Q10: I prefer learning with this app over other ways.	Q1: My child/student was motivated to use the app. Q3: The app captured and maintained their attention during the session. Q10: I would like them to continue using the app regularly.
Independence	Q5: I could use the app without an adult's help. Q7: I can remember the steps without looking at the app.	Q7: They have shown more independence when performing the task after using the app.
Global score	Average of Q2–Q10 (without Q1)	Average of Q1–Q10

Table 4: Distribution of questions according to the evaluated dimensions.

The table above shows the distribution of questions according to the measured dimensions, as well as their grouping into indices used for statistical analysis. For children, it is important to note that question Q1, related to the use of the VR headset, was excluded, since no child used this tool. This decision aimed to avoid biasing the results. For parents, all ten questions were retained in the calculation of the indices.

These groupings form the basis of the subsequent analysis, allowing the examination of the different dimensions, the internal consistency of responses, as well as the correlations between the perceptions of children and those of their parents.

Children	Mean	SD
Usability	3.30	1.49
Learning	2.95	1.23
Engagement	4.28	0.99
Independence	3.32	1.67
Overall	3.36	1.18
Parents	Mean	SD
Usability	3.45	1.24
Learning	3.44	0.89
Engagement	3.32	0.62
Independence	2.55	1.10
Overall	3.34	0.66

Table 5: Mean scores and standard deviations ($\pm$ SD) for children and parents

The study was conducted with 20 child–parent dyads ($N = 20$), which provides a comparative view of perceptions across the two groups. The analysis of means and standard deviations ($\pm$ SD) highlights several tendencies in how children and parents evaluated the intervention.

Children reported a high level of engagement ($M = 4.28$, $SD = 0.99$), suggesting that the activity was perceived as enjoyable and motivating for most participants. Usability ($M = 3.30$, $SD = 1.49$) and independence ($M = 3.32$, $SD = 1.67$) received more moderate ratings, accompanied by relatively high variability, reflecting notable differences across children depending on their abilities or prior experiences. Perceived learning ($M = 2.95$, $SD = 1.23$) was somewhat lower, indicating that the educational impact of the tool was more variable. The overall child score ($M = 3.36$, $SD = 1.18$) was at a medium level, reflecting a generally positive but nuanced evaluation.

Parents rated usability ($M = 3.45$, $SD = 1.24$) and learning ($M = 3.44$, $SD = 0.89$) favorably. Engagement ($M = 3.32$, $SD = 0.62$) was also evaluated positively, with lower variability across respondents (smaller SD). However, independence ($M = 2.55$, $SD = 1.10$) was perceived as relatively low, indicating that parents considered their children to require substantial assistance when using the tool. The overall parental score ($M = 3.34$, $SD = 0.66$) was very close to the children's score, reflecting a consistent overall perception between the two groups.

In summary, children expressed strong enthusiasm and engagement, whereas parents placed more emphasis on usability and learning benefits. The most notable discrepancy between groups concerned the perception of independence, with parents being more critical than children about the degree of autonomy achieved.

Interpretation of the Graph Usability

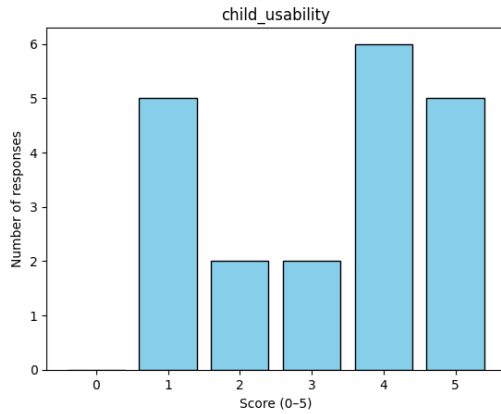


Figure 20: Child Usability

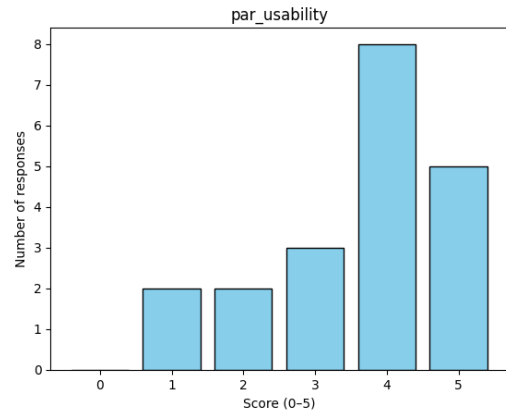


Figure 21: Parent Usability

Both children and parents generally reported positive perceptions of usability, although some differences emerged.

Children: responses were more dispersed, ranging from low (1–2) to very high (4–5). While several children encountered difficulties, a majority assigned higher scores, indicating that they considered the tool relatively easy to use.

Parents: responses were more homogeneous and concentrated at the upper end of the scale (scores 4–5). This suggests that parents perceived the tool as straightforward and effective in terms of usability.

Comparison: children's evaluations revealed greater variability, whereas parents were more consistent in considering the tool highly usable.

Interpretation of the Graph Learning

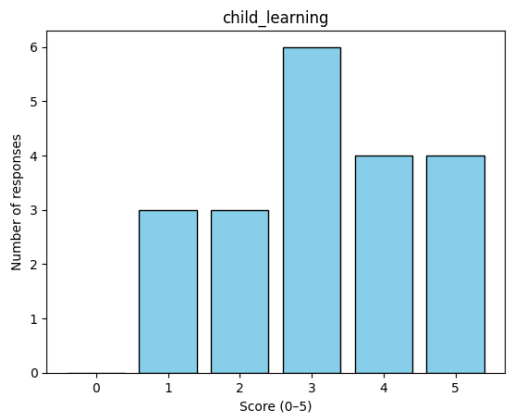


Figure 22: Child Learning

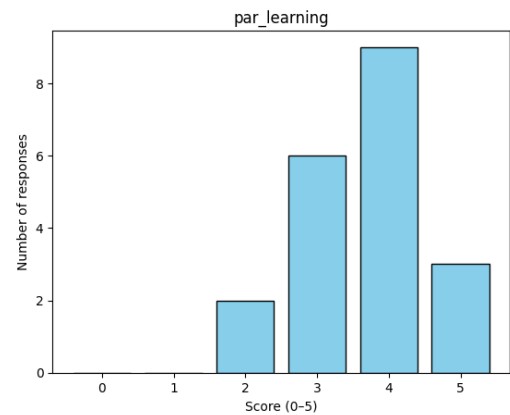


Figure 23: Parent Learning

Children and parents diverged somewhat in their assessments of learning outcomes.

Children: responses were centered around the mid-range (score 3), reflecting moderate perceived learning benefits. Some children rated the learning experience as low (1–2), while others gave high scores (4–5), highlighting heterogeneous perceptions.

Parents: the majority of parents' ratings clustered around 3 and 4, reflecting stronger confidence in the tool's educational value.

Comparison: parents rated the learning impact more positively, whereas children's evaluations were more mixed, suggesting a less clear perception of learning gains.

Interpretation of the Graph Engagement

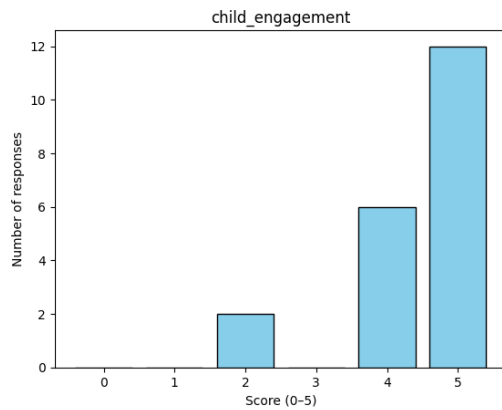


Figure 24: Child Engagement

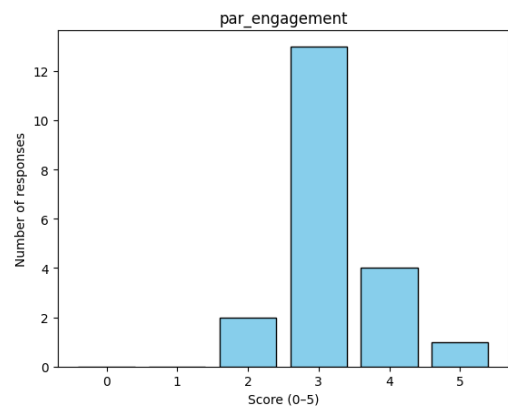


Figure 25: Parent Engagement

Engagement revealed the strongest discrepancy between children and parents.

Children: responses were overwhelmingly positive, with the majority assigning the highest scores (4–5). This indicates that children were highly motivated and emotionally involved in the activity.

Parents: responses were concentrated around the mid-level (score 3), with fewer high ratings. Parents recognized a moderate level of engagement, but not as intense as reported by the children.

Comparison: children expressed strong enthusiasm and immersion, while parents offered more cautious and tempered evaluations of engagement.

Interpretation of the Graph Independence

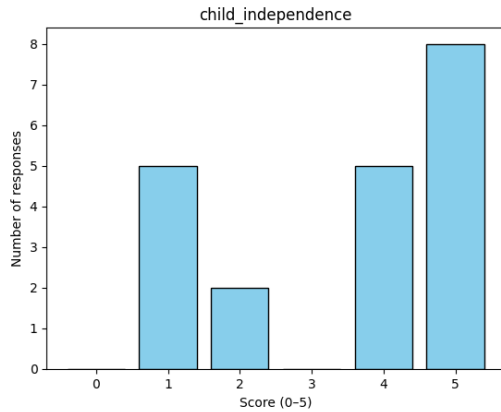


Figure 26: Child Independence

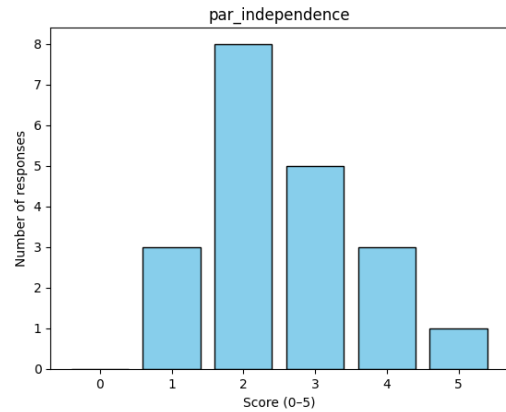


Figure 27: Parent Independence

The dimension of independence revealed notable divergences.

Children: responses were polarized, with some children reporting low autonomy (scores 1–2) while others considered themselves highly independent (scores 4–5). This variability suggests that autonomy was not experienced consistently.

Parents: the majority assigned low to moderate scores (2–3), indicating that they perceived children as still requiring substantial support. Very few parents considered their children to be fully independent.

Comparison: while some children self-reported high independence, parents generally remained skeptical about their ability to work autonomously.

Overall Evaluation

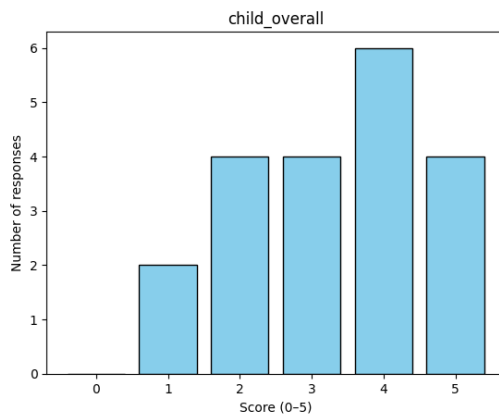


Figure 28: Child Overall

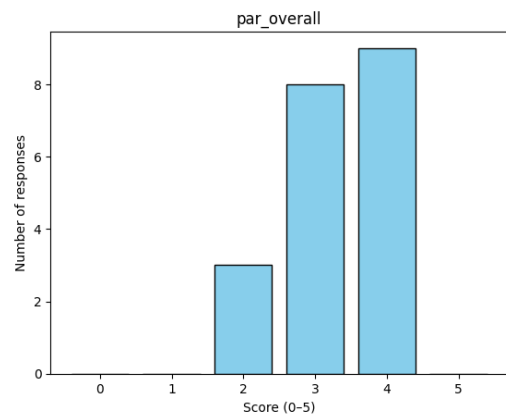


Figure 29: Parent Overall

Children: overall responses ranged widely but tended toward positive values, with many scores concentrated around 4. This reflects a favorable yet varied overall perception.

Parents: responses were more uniform, concentrated around scores 3 and 4, indicating a generally positive but more measured evaluation.

Comparison: children expressed more diverse opinions, whereas parents adopted a more moderate and consistent stance in their global evaluation.

Internal Consistency of the Questionnaires (Cronbach's Alpha)

The internal reliability of the questionnaires was assessed using Cronbach's Alpha. For the children's questionnaire, the alpha coefficient was 0.93, indicating excellent internal consistency. This suggests that the ten items included in the children's scale were highly homogeneous and measured the same underlying construct. The strong inter-item correlations imply that the instrument was very reliable for capturing children's perceptions in this study.

For the parents' questionnaire, Cronbach's Alpha was 0.75, which represents acceptable to good reliability according to psychometric standards. Although lower than the value obtained for the children's responses, this score indicates that the parent scale was sufficiently consistent to justify its use in analysis. Some variability among items is present, but overall the scale remains coherent.

In summary, both questionnaires demonstrate adequate internal consistency, with particularly high reliability for the children's responses. These findings reinforce the robustness of the data and support the validity of the subsequent analyses based on composite scores.

Child–Parent correlation (overall score)

The analysis shows a very strong positive correlation ($r = 0.88$) between children's and parents' overall scores, suggesting that both groups tended to provide similar evaluations. The reported p-value ($p = 0.000$) indicates statistical significance under conventional thresholds. However, it is important to note that the study is based on a relatively small sample size ($N = 20$ dyads). In such cases, the reliability of the p-value may be limited, and the result should be interpreted with caution. The high correlation coefficient remains meaningful, but further studies with larger samples are necessary to confirm these findings with greater confidence.

Mean Overall Scores by Disorder

This study, based on a sample of 20 child–parent dyads, examined the average overall scores across different neurodevelopmental disorders. The findings highlight notable variations between groups, both from the children's and the parents' perspectives.

Children Among children, overall scores showed substantial heterogeneity depending on the reported disorder. The highest values were observed in children with Dysphasia ($M = 4.78$) and those diagnosed with ADHA ($M = 4.89$), reflecting a generally positive perception of the proposed activity. Children with ADHD ($M = 3.93$) and dyslexic difficulties ($M = 3.86$) reported intermediate scores. By contrast, children with anxiety disorder (AD, $M = 3.33$) and especially those with autism spectrum disorder (ASD, $M = 2.70$) showed the lowest scores, suggesting a less favorable perceived experience.

Parents Parental evaluations followed a similar overall trend, though they were less polarized. The highest mean scores were reported by parents of children with Dysphasia ($M = 4.10$) and ADHA ($M = 3.80$). Parents of children with ADHD ($M = 3.47$), dyslexic difficulties ($M = 3.55$), or anxiety disorder (AD, $M = 3.70$) indicated mid-range values. Finally, the lowest mean scores were assigned by parents of children with ASD ($M = 3.07$), confirming, at least partially, the perception expressed by the children themselves.

Comparative Analysis A comparison of child and parent evaluations reveals a general alignment in trends, although children tend to express more extreme judgments (either higher or lower). This divergence may reflect differences in perspective: children primarily evaluate their immediate subjective experience with the activity, whereas parents adopt a more balanced view, integrating broader considerations related to the child's overall behavior and developmental context.

7.4 General conclusion

This study aimed to design, develop, and evaluate a digital tool to support children with neurodevelopmental disorders (NDD) in their daily routines. The findings obtained from a

sample of 20 child–parent dyads suggest that the application provides a solid and promising foundation for enhancing children’s engagement, learning, and independence, while also being perceived positively by parents in terms of overall usability.

Statistical analyses revealed a high level of internal consistency for the children’s items ($\alpha = 0.93$) and satisfactory reliability for the parents’ items ($\alpha = 0.75$), confirming the robustness of the scales employed. Moreover, the strong overall correlation between child and parent evaluations ($r = 0.88$) indicates convergence in perceptions regarding the usefulness of the tool. However, the relatively small sample size limits the scope of statistical inference, particularly with respect to the reliability of p-values.

Overall, the results highlight the relevance of adopting a transdiagnostic and ecological approach. The application shows considerable potential in supporting the development of executive functions in children. Nonetheless, further improvements are needed to maximize its effectiveness and ensure sustainable transfer into real-life contexts. Potential avenues for future development include the integration of an immediate and positive feedback system, the implementation of a more predictable task structure, and the expansion of available missions to cover a broader range of daily routines.

In conclusion, this research provides a valuable foundation for future work aimed at refining and enriching the tool, while reinforcing its scientific and clinical significance.

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